

United International University

B.Sc. in Computer Science & Engineering (CSE)

CSE 4889: Machine Learning

Midterm Exam: Spring 2026 Time: 1 Hour 30 Minutes Marks: 30

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Answer all of the following questions sequentially.

1. In Fuzzy C-means clustering, the membership degree u_{ij} of a data point x_j belonging to cluster i is calculated using the following objective function derivative: [6]

$$u_{ij} = \frac{1}{\sum_{k=1}^C \left(\frac{\|x_j - v_i\|}{\|x_j - v_k\|} \right)^{\frac{2}{m-1}}}$$

where:

- u_{ij} is the degree of membership of x_j in the cluster j
- $\|x_j - v_i\|$ is the distance between point x_j and cluster center v_i .
- C is the total number of clusters.
- m is the fuzziness coefficient ($m \in [1, \infty]$).

Based on the formula above, mathematically explain what happens to the membership values u_{ij} if the value of m is set exactly to 1.

2. a) Fit the Random Forest Regression model to predict $\mathbf{B}$ based on $\mathbf{A}$ where each decision tree (T) is trained on minimum four different subsets generated from the following dataset. Moreover, the final prediction is defined as follows: [4]

$$\hat{B} = \frac{1}{T} \sum_{i=1}^T f_i(A)$$

A	3	5	7	11	15	23
B	8	12	16	20	24	32

- b) HDBSCAN transforms the data space using the Mutual Reachability Distance. [4]
Explain how this specific distance metric helps the algorithm handle clusters of varying densities more effectively than standard DBSCAN.

3. Consider the following set of two-dimensional data points: [2+4+2]

D1(2, 3), D2(3, 4), D3(3, 5), D4(6,7), D5(1, 2), and D6(4, 5).

Using an appropriate clustering approach and distance metric:

- Construct the pairwise distance matrix for the data points.
- Show the step-by-step merging process and draw the corresponding dendrogram for the dataset.
- Based on the dendrogram, determine the most appropriate number of clusters and justify your answer.

4. A dataset of products sold in different super shops is given as follows: [8]

Product	Shop	Prices	VAT	Discount
Apple juice	Shwapno	600	95	7.0
Apple juice	Shwapno	650	87	8.5
Ice cream	Unimart	400	86	9.0
Ice cream	Unimart	425	88	10.0
Coffee	Agora	2110	110	11.0
Coffee	Agora	452	105	10.5

You are asked to train a Random Forest Classifier (with resampling where each tree uses at least 3 samples, i.e., $S \geq 3$) to classify products based on the features Shop, Prices, VAT, and Discount.

After training, classify the following unknown samples:

Product	Shop	Prices	VAT	Discount
?	Shwapno	425.5	46.5	3.5
?	Unimart	1300	120	8.5
?	Agora	400	50	5.0